The Late Devonian Paleomagnetic Field for Southeastern Australia: A New Result for the Lochiel Formation and a Reassessment of Results From the Catombal Group

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Paleomagnetic results from a new investigation of the Upper Devonian Lochiel Formation from southeastern Australia contrast with results obtained previously. Five flow units yielded a mean direction of magnetic remanence, stable to alternating field treatment, with declination $D = 331°$ and inclination $I = 10°$. Secondary components of magnetization acquired approximately parallel to the mid-Carboniferous field and the present field were retained after alternating field treatment on specimens from two more flow units. These results are compared with results obtained by the vector reduction technique for isolating primary components from the Upper Devonian Catombal Group. Reassessment of those data shows that corroboration of the earlier results for the Lochiel Formation was fortuitous; they, in fact, support results obtained through this new study. Preliminary observations of the Lochiel volcanics and the overlying Merrimbula Group sediments suggest that they are extensively deformed into zigzag monoclinal folds. The dispersion that remained after correction for bedding tilt can be attributed to an incomplete understanding of the deformational phases in the region.

INTRODUCTION

New paleomagnetic results for the volcanic horizons of the Upper Devonian Lochiel Formation of southeastern Australia have indicated the presence of structural complexities which bring into question the significance of some previously reported data [*Luck*, 1973]. Furthermore, it has been discovered that support for *Luck*'s [1973] data from results reported by *Williamson and Robertson* [1976] was in fact fortuitous and was based on an incorrect value of the dipole field direction at the collecting sites in the Catombal Formation. The true value of the dipole field used in their iterative method gives a direction for the Late Devonian field that is different from Luck's.

During a field sampling program designed to extend our work on the Devonian of Australia, we sampled new localities in the volcanic horizons of the Lochiel Formation. Luck followed *Green* [1961] and restricted his sampling to outcrops south of Nethercote; here, we sampled further north between Lochiel and Nethercote. Although ours is rather a small study, the results indicate that the Lochiel Formation is strongly deformed and the variable attitudes of the flows may result from the interactions of two fold phases. Reconstructions of fold geometry are hindered by poor exposure and the paucity of way up criteria. Data obtained without a proper structural analysis being carried out to make bedding corrections for postmagnetization folding may be considered suspect.

This new result for the Upper Devonian Lochiel Formation differs from *Luck*'s [1973] result but does support the result obtained by using the corrected dipole field direction in *Williamson and Robertson*'s [1976] iterative method for isolating the primary component of remanent magnetization from sediments of the Upper Devonian Catombal Group.

SOME STRUCTURAL OBSERVATIONS

A preliminary traverse was made across the more southerly outcrops sampled by *Luck* [1973]. That region would appear to exhibit a structural pattern rather more complex than was indicated by the evidence that was then available, viz., 'The rocks have been gently folded along meridional axes, with a slight regional tilting to the east.' However, Luck also noted that at one locality 'the lack of between sample agreement probably results from structural deformation . . . which represents a zone of weakness in the formation.'

To date there has been no detailed structural analysis carried out in the Nethercote-Pambula area. Our observations at the outcrops which Luck sampled indicate that major zigzag monoclinal folds deforming the volcanics do occur, with wavelengths up to about 500 m and with axes which plunge at 10° to an azimuth of 020° (true north). The folds have acute, narrow hinge zones with limbs which vary in dip between $<10°$ and 90°, and the layering has a constant thickness throughout the folds. They are tentatively assigned to be subclass 1B type parallel folds [*Ramsay*, 1967, pp. 367, 436].

Well-exposed new road sections between Nethercote and Eden (37.05°S, 149.85°E) in the conformably overlying Upper Devonian Merrimbula Formation clearly demonstrate the prevalence of zigzag monoclinal folding in the region. *Hobbs* [1962], *Williams* [1971], and *Hobbs et al.* [1976, p. 172] have described angular, asymmetrical style folds from coastal sections in the Bermagui and Narooma region approximately 35 km to the north of Merrimbula. In particular, *Williams* [1971] assigned these folds to a second phase (B_2) of deformation and described an earlier (B_1), tighter set of folds which 'are commonly recumbent' and refolded by B_2 structures. South of Nethercote we could find no evidence of B_1 tight folds, but the sporadic outcrops virtually preclude identifying such structures because of the lack of available features in the volcanics suitable for use as younging criteria. The monoclinal folds around Nethercote therefore may correlate with the B_2 folds of Williams. It is therefore not surprising that without the availability of a detailed structural map, apparently conflicting paleomagnetic results have been obtained.

PALEOMAGNETIC RESULTS

The Lochiel Formation was sampled at 36°56'S, 149°46'E (LV1 and LV2) and over an outcrop distance from 36°57'S to 37°00'S along 149°49'E. Samples from seven flow units (sites) were cored in the field with a hand-held power drill. From measurements made of the flow banding orientation, two sites (LV1 and LV2) were located on a northeasterly dipping fold limb; the flow unit at site LV3, from observations of clearly

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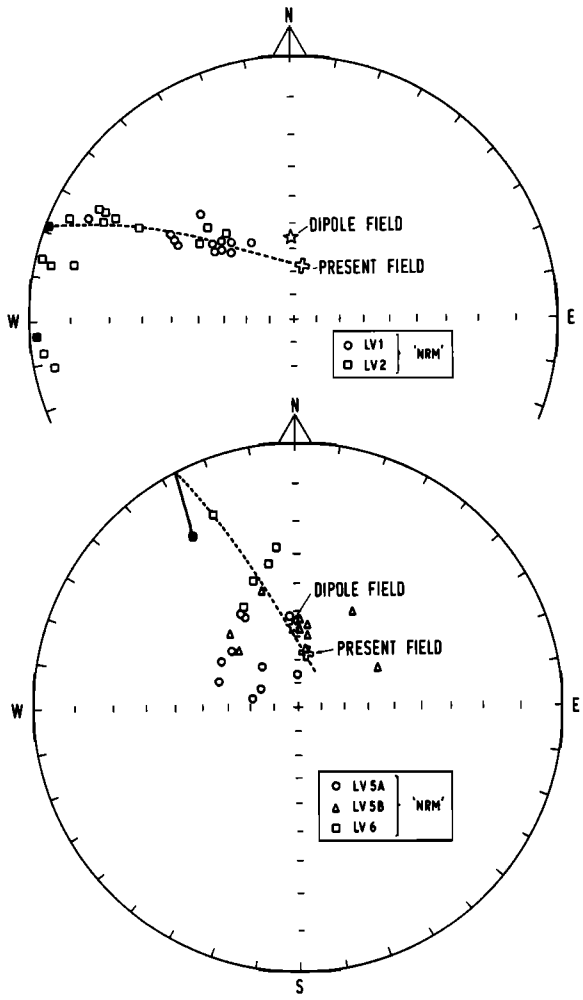


Fig. 1. Distribution of natural remanent magnetization directions from flow units LV1, LV2, LV5A, LV5B, and LV6 from the volcanic horizons of the Lochiel Formation. Open symbols denote negative upward and solid symbols positive downward pointing directions plotted on a Wulff equal angle stereographic net.

visible aligned fragments, was dipping westward, and flows LV4, LV5A, LV5B, and LV6 exhibited massive layering, indicating a northwesterly dip. The lack of further detailed structural information is largely due to the limited nature of the outcrop.

Directions of natural remanent magnetization (NRM) were measured with a complete results Digico magnetometer. Initially, directions either clustered around the present magnetic field direction at the sampling locality, showed some evidence of a planar distribution away from the present field (some examples are shown in Figure 1), or, in the case of LV3, were scattered around the stereogram, apparently lacking any preferred orientation. NRM intensities of magnetization exhibited little variation, being typically within the range 10–50 mA m^{-1} (= 1×10^{-5} to 5×10^{-5} G), with the exception of flow unit LV4, whose NRM intensities were consistently an order of magnitude greater, lying within the range 300–700 mA m^{-1}.

One specimen from each core was partially demagnetized by using a Schonstedt GSD-1 unit at 10-mT (= 100 Oe) field increments to a maximum of 60 mT. The relative stability of magnetization to alternating field (af) treatment was visually estimated from an inspection of the coercive force spectrum. Directions of magnetization for specimens from flow units LV1, LV2, LV5A, LV5B, and LV6 changed systematically along the great circles defined approximately from the distributed NRM results. Generally, little change was observed between the 40-mT and 60-mT treatments (see Figure 2*a*): it was considered that treatment at 50-mT af was adequate to isolate the hardest component of magnetization. The remainder of the specimens from those sites were cleaned at 50 mT; consistency was checked by also cleaning at 60 mT. A summary of the results is listed in Table 1, and from Figure 2*b* it can be seen that directions of magnetization, whether plotted with respect to the present horizontal or the bedding (the assumed paleohorizontal), are strongly oblique to the present magnetic and dipole field axes. In the case of these five flow units, correction for the apparent tilt constitutes an application of the fold test [*Graham*, 1949]. The precision parameter k, defined by *Fisher* [1953], increases by a factor of 2.41. According to *McElhinny* [1964], the 95% confidence limit for the application of the fold test, for $N = 5$, is 3.44. Although the scatter is reduced upon correction for structural tilting, the fold test is not significant at the 95% confidence level.

Secondary Magnetizations and Vector Reduction

Remanence directions in specimens from flow unit LV4 moved systematically on cleaning from a position near the present field, but only to occupy a stable position slightly oblique to it (Figures 2*c* and 2*d*). However, directions from specimens of flow unit LV3, which were initially strongly oblique to the present field, clustered very convincingly around the present magnetic field axis after treatment up to 60 mT (also shown in Figure 2*d*). Clearly, these two flows have acquired hard secondary components of magnetization that could not be removed preferentially. The significance of the secondary directions was tested with respect to known field directions which plot in similar positions. The mean direction of flow LV3 is not different from the present magnetic field direction (013°, −67°, as estimated from the International Geomagnetic Reference Field 1965.0 [*Leaton and Barraclough*, 1971]), though it is different from the axial dipole field (000°, −56.4°). The direction of flow unit LV4 is significantly different from both field directions. Interestingly, it is not different

TABLE 1. Summary of the Primary Directions of af-Cleaned Remanent Magnetization

Flow Unit	N*	Present Horizontal,† deg		α_{95},‡ deg	Bedding Corrected,† deg	
		D	I		D	I
LV1	12	292.0	−11.5	9.0	290.0	+2.0
LV2	16	288.0	+1.5	5.5	291.0	+15.5
LV5A	8	330.0	+41.0	13.5	322.5	+18.0
LV5B	8	327.0	+27.5	12.0	323.0	+5.0
LV6	6	329.5	+30.0	12.5	325.0	+7.5
Mean of flow means	5			18.5	311	+10

Pole position calculated from flow VGP's is 28°N, 092°E; A_{95} = 17.5°.

*The value N is the number of specimens per flow unit (average of two per core).

†D and I are the declination and inclination of magnetization, respectively.

‡The value α_{95} (A_{95}) is the semiangle of the cone of confidence calculated according to *Fisher* [1953] at the 95% probability level.

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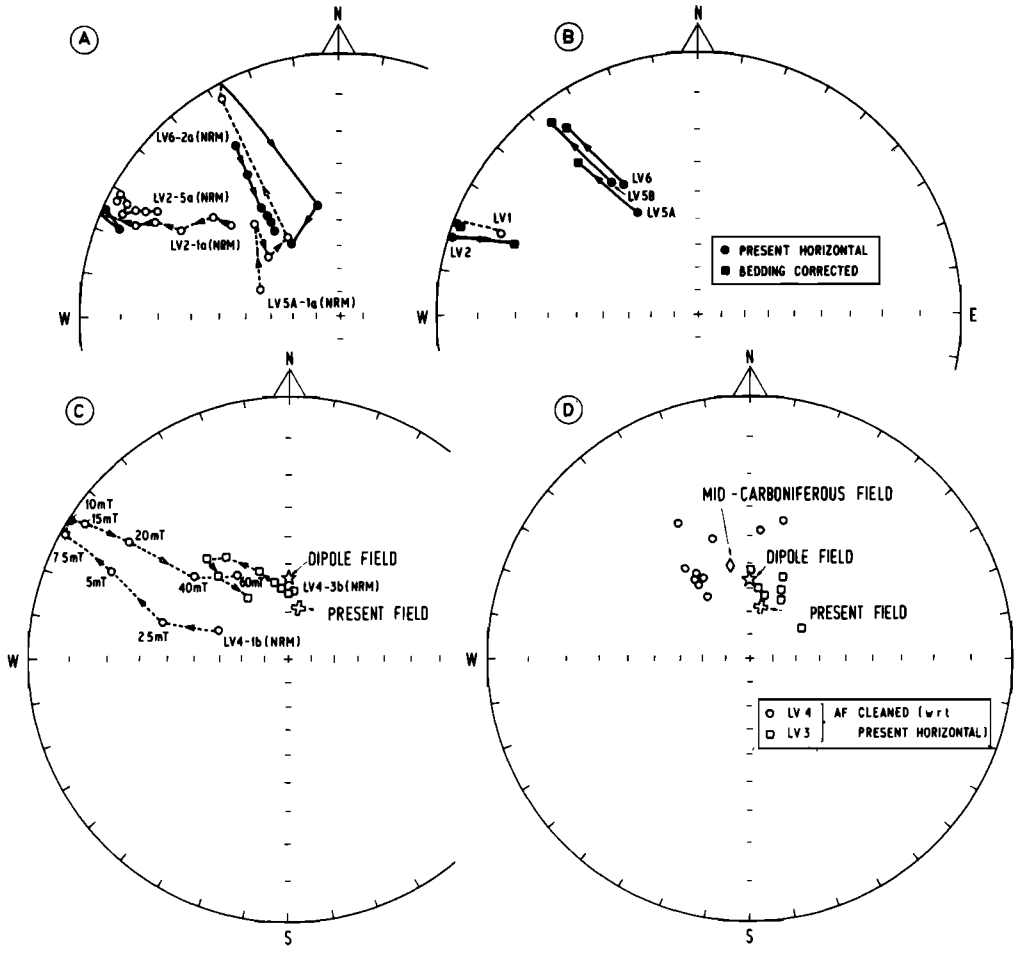


Fig. 2. The af-cleaned directions of remanent magnetization. (*a*) Examples of the response to stepwise af treatment at 10-mT field increments to a peak of 60 mT. (*b*) Mean directions of flow units considered to have retained a component of primary magnetic remanence. (*c*) Detailed af treatment of specimens from flow unit LV4 has revealed a complex multicomponent magnetization. The component with the highest coercive force is only slightly oblique to the dipole and present magnetic field axes. (*d*) Cleaned specimen directions for flow units LV3 and LV4 shown in relation to the present, dipole, and mid-Carboniferous magnetic fields. Open symbols denote negative upward and solid symbols positive downward pointing directions plotted on a Wulff equal angle stereographic net.

from the mean mid-Carboniferous field direction (349°, −51°) calculated from the Gilmore and Isismurra volcanics [*Irving*, 1966; *Luck*, 1973] and the Yetholme adamellite [*Facer*, 1976]. It is not unreasonable to assume that the interval of time around the mid-Carboniferous was a period during which secondary magnetizations may have been acquired: the early Carboniferous was a period of orogeny (the Kanimblan) in southeastern Australia.

Comparison With Previous Results

Williamson and Robertson [1976] have provided us with a useful analytical technique for stripping off secondary components of magnetization. Application of the technique was illustrated by using samples of the Upper Devonian Catombal Group of Australia, collected in a locality where a suitable spectrum of structural attitudes was reportedly available. A number of possible approximate primary directions were obtained by vector reduction along a selection of directions of the earth's magnetic field assumed to have prevailed in that locality at some time subsequent to the acquisition of the primary magnetization. Directions tested therefore were late Carboniferous (declination D = 023° and inclination I = −78°, although the magnetization of the Percy Creek volcanics [*Chamalaun*, 1968] and the Upper Kutting sediments [*Irving*, 1966] dates from the late Paleozoic reversal interval; it may therefore have been more realistic to attempt a vector reduction along the reversed direction to test this possibility fully), Jurassic, and a number of points intermediate between the Jurassic field direction and the present dipole field. However, the secondary component reduced along the direction 000°, −42° was mistakenly reported as that representing the present dipole field.

The collecting locality near Wellington, New South Wales, is situated at 33°S, and, according to the dipole formula, tan I = 2 tan λ, where I is the inclination of magnetization and λ is the geomagnetic/geographic latitude, the normal direction of the dipole field axis lies at 000°, −52.4°. As was stated by *Williamson and Robertson* [1976] and clearly shown in their Table 2, '. . . quite small differences in choice of secondary direction yield quite large differences in the resulting primary direction and scatter.' Although reduction along the direction 000°, −42° produced a population with the minimum dispersion, the physical significance of this direction is obscure, and its use has led to the calculation of an erroneous result. Unfortunately, it was used to corroborate a result reported by *Luck* [1973] for the Lochiel Formation of southeastern Australia.

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The question that arises therefore is, in order to employ the vector reduction technique of *Williamson and Robertson* [1976], which direction can be most reliably chosen as the secondary magnetization for a simple two-component system? In the case of the Lochiel Formation, either the mid-Carboniferous or the present magnetic field direction is an obvious choice. The present axial dipole field is less likely to have been a significant influence during the acquisition of a viscous component of remanent magnetization. The steeper present field is the preferred choice from the experimental results presented here. Flow unit LV4, which probably acquired a hard secondary magnetization during the mid-Carboniferous, has also acquired a softer viscous component along the present field direction: its magnetization is therefore complex and multicomponent.

Of course, direct applicability of the results reported here to another formation, even one of the same age, may be regarded as a tenuous extrapolation of the data. However, some experimental evidence is now available which allows us to predict more confidently possible remagnetization axes for use in conjunction with the vector reduction technique: it must also be demonstrated that the magnetization is two component. Clearly, a direction steeper than that used originally by Williamson and Robertson must be used if we are to relate the secondary component to the axial dipole field (i.e., $I = -52°$, not $-42°$). The inclination of the present field at Wellington is about $-63°$, the mid-Carboniferous field is $\sim -51°$, and the dipole field is $\sim -52°$. From their Table 2, estimates of primary directions obtained after vector reduction using the secondary components with inclinations $-62°$ and $-52°$ are not unlike the direction obtained here by experimental procedure from the Lochiel Formation. The principal difference between the two results is in the definition of polarity.

Discussion

The mean direction of magnetization obtained from this study, viz., a declination $D = 311°$ and an inclination $I = 10°$, is different from the previously reported result ($D = 355°$, $I = -11°$ [*Luck*, 1973]) for the volcanic horizons in the Lochiel Formation. A careful attempt was made in the present study to obtain the relevant structural information at each collecting site to enable structural unfolding, but a considerable amount of scatter was still evident in the final results (although the precision parameter k increased by a factor of 2.41). There are three main possibilities to consider regarding the persistence of the scatter: (1) a large component of secular variation is present, (2) different sites (limbs) represent different points in time between which the ambient field has changed direction (unlikely except as part of (1)), and (3) the structural information remains incomplete. The last is the most likely, since the rock exposures were not adequate to determine the structure completely.

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